

Case 1: Pharmacy Cold Chain to Inpatient Floor (IPD)

The Story in Plain English

A patient in an inpatient hospital room needs a delicate, temperature-sensitive medication (stored in a refrigerator). A floor nurse submits a digital request (an indent) for the drug. The pharmacy software must read the doctor's official prescription to double-check that the drug and dose are right, package the medicine, update the patient's digital chart when the courier leaves, and send a notification to the nurse's phone showing who is bringing it and when it will arrive—without leaking private patient data on the notification screen.

What the Candidate Must Build

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- **Read the Prescription:** Query the EHR using the [HL7 FHIR MedicationRequest Specification](#) to inspect the doctor's original order.
- **Validate the Drug:** Check the medication name against the official [NIH NLM RxNav / RxNorm API](#) to confirm that the requested drug formulation matches the prescription.
- **Process the Delivery Event:** Parse an incoming legacy hospital message formatted as an HL7 v2 order string (OMP^O09) using an open-source parsing library.
- **Update the Chart:** Write a new record back to the system using the [HL7 FHIR MedicationDispense Specification](#) to log that the drug was packed, who is carrying it, and its ETA.
- **HIPAA Compliance:** Strip all identifying PHI from the outbound alert sent to the nurse's pager/app, and write a tamper-proof audit trail using the [HL7 FHIR AuditEvent Specification](#).
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Official Open-Source Tools & Sandboxes for Candidates

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- **FHIR Server Sandbox:** Candidates can run a local test server using the official open-source [HAPI FHIR JPA Server Starter on GitHub](#) or query the public [HAPI FHIR Public Test Sandbox](#).
- **HL7 v2 Message Parsing Library:** Candidates can use the [Redox HL7 v2 Parser on GitHub](#) (Node.js/TypeScript) or the [HL7apy Library on GitHub](#) (Python). [7]

Case 2: Operating Theater (OT) Pre-Surgical Safety Gate

The Story in Plain English

Before a surgeon cuts into a patient, the operating room team runs through a safety checklist. Instead of using paper clipboards, the team opens an app inside their EHR monitor. The app must automatically look up the scheduled surgery, confirm that the patient signed the consent for that specific procedure, check that the patient isn't allergic to any surgical antibiotics, and check that recent blood clotting tests look safe. Finally, it compiles an official surgical summary document for the record.

What the Candidate Must Build

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- **Launch Inside the EHR:** Build a web screen that uses the official [SMART on FHIR App Launch Framework](#) to log in automatically and identify which patient is currently on the operating table.
- **Verify Diagnosis and Surgery:** Cross-check the scheduled procedure using official procedure terminology ([CPT](#)) against the patient's diagnosed condition using [SNOMED CT Browser](#).
- **Check Safety Labs & Allergies:** Fetch blood lab panels (like platelet count or clotting times) from the EHR using the [HL7 FHIR Observation Specification](#) by filtering specifically on standardized [LOINC Terminology Codes](#), and check allergy risks via the [HL7 FHIR AllergyIntolerance Specification](#).
- **Standardized Document Export:** Package the verified pre-op checklist into an industry-standard clinical note structure following the federally required USCDI Standards via HealthIT.gov. [3]
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Official Open-Source Tools & Sandboxes for Candidates

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- **SMART App Simulator:** Candidates can simulate an EHR interface and test their login flow using the official SMART on FHIR App Launcher.
- **EHR Embedding Client:** Front-end developers can use the open-source [SMART on FHIR Client JS Library on GitHub](#).
- **Clinical Document Reference:** Guidelines and schemas can be referenced directly from the [HL7 CDA Core Repository on GitHub](#). [8]

Case 3: Inpatient Chemotherapy Infusion Safety Gate

The Story in Plain English

Chemotherapy drugs are extremely powerful and can be toxic if given at the wrong time or at incorrect doses. When an inpatient cancer patient is due for an infusion, an automated safety-gate service must pull the oncologist's multi-drug prescription, check this morning's lab results to ensure the patient's white blood cells and kidneys can handle the dose, and update the bed monitor when the pharmacy releases the batch. When the treatment ends, the software bundles the record so it can be transmitted safely to outside cancer registries.

What the Candidate Must Build

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- **Parse Complex Drug Regimens:** Fetch the oncology protocol using the HL7 FHIR MedicationRequest Specification, writing logic to handle complex nested arrays that specify multiple mixed ingredients, carrier fluids, and exact drip rates.
- **Automated Lab Safety Gate:** Programmatically retrieve the patient's latest Absolute Neutrophil Count (ANC) and Serum Creatinine levels using specific LOINC Lab Codes. If the numbers fall below safe thresholds, the code must raise an automated hold flag.
- **Receive Floor Administration Alerts:** Ingest an older hospital bedside message formatted as an HL7 v2 administration event (RAS^O17) to record when the IV pump starts running.
- **Network Interoperability & Legal Trust:** Ensure that data saved downstream complies with a simulated **Business Associate Agreement (BAA)** policy, and structure the discharge data payload to be compliant with national data sharing rules under TECCA via HealthIT.gov. [9]
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Official Open-Source Tools & Sandboxes for Candidates

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- **Drug Terminology Engine:** Candidates can fetch active ingredients and brand relationships programmatically with the free, official [NLM RxNav REST API Documentation](#).
- **Server & Data Validation:** Use the [HL7 FHIR Validator CLI on GitHub](#) to verify that complex oncology JSON payloads conform strictly to official FHIR schemas. [10]
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Common Mistakes

Criteria	 Red Flags ("Novice Vibe Coder")	 Green Flags (True Health Vibe Coder)
Data Querying	Uses string matching (e.g., if (name.includes("Blood Pressure")))	Uses standard LOINC or RxNorm code lookups on API endpoints
Legacy HL7 Parsing	Splits raw strings with manual regex or `text.split('`	')
Authentication	Stores cleartext tokens in localStorage or hardcodes IDs	Uses PKCE OAuth 2.0 via SMART on FHIR launch context
Data Privacy (HIPAA)	Leaves names, phone numbers, or dates of birth in public alert payloads	Strips unneeded PHI , honors data minimization, and writes AuditEvent records

[1] <https://lhncbc.nlm.nih.gov>

[2] <https://www.hl7.fi>

[3] <https://build.fhir.org>

[4] <https://hl7.org>

[5] <https://www.nlm.nih.gov>

[6] <https://hl7.org>

[7] <https://github.com>

[8] <https://docs.smarthealthit.org>

[9] <https://www.cdc.gov>

[10] <https://lhncbc.nlm.nih.gov>

Key Terms Explained Simply

- **EHR (Electronic Health Record):** The core software system hospitals use to store patient charts, diagnoses, and orders (like Epic, Cerner, or an open-source clone).
- **FHIR (Fast Healthcare Interoperability Resources):** A modern set of rules and JSON formats that allows software to read and write hospital records using standard web APIs (HTTP GET/POST), just like Twitter or Stripe APIs.
- **HL7 v2:** An older, 1980s text-message format (separated by pipe | characters) still used inside almost every hospital for real-time notifications, like when a patient checks into a room or pharmacy orders are sent.
- **SMART on FHIR:** A secure login workflow (OAuth 2.0) that lets a custom web app launch directly inside a doctor's EHR screen without having to log in separately.
- **RxNorm:** A standardized catalog of medication names, strengths, and ingredients run by the US National Library of Medicine (NLM) so computers never confuse brand names with generic chemical formulas.
- **LOINC:** A universal database of ID numbers for laboratory tests and vital signs (e.g., one specific code always means "Systolic Blood Pressure," regardless of which hospital took the reading).
- **SNOMED-CT & CPT:** Standard code lists used for clinical conditions/diseases (SNOMED) and billable medical/surgical procedures (CPT).
- **HIPAA & PHI: Protected Health Information (PHI)** is any private data that can identify a patient (name, birth date, phone number). **HIPAA** is the US law that imposes heavy fines if software leaks PHI or stores it insecurely.
- **BAA (Business Associate Agreement):** A legally binding contract required under HIPAA when an outside developer or software company handles hospital patient data.
- **USCDI (US Core Data for Interoperability):** A federally mandated checklist of minimum health data classes (allergies, vital signs, notes) that every health app must support sharing.
- **HIE & TEFCA:** An **HIE (Health Information Exchange)** is a network allowing different regional hospitals to share records. **TEFCA** is a nationwide government agreement connecting all these regional networks together into one trusted pipeline. [1, 2, 3, 4, 5, 6]